Recipe 4: Optimizing Complex Database Queries

It looks like your app is having some performance issues when it comes to retrieving posts and comments. You figured out that the N+1 query problem was behind the slowdown. This issue comes up when you fetch all the posts and then loop through each one to get its comments, which makes an extra query for every post. To solve this in our Laravel 11 project running on Linux,we will use Laravel's built-in eager loading feature. Now, we will learn how to improve your data retrieval by using eager loading, a specific method provided by Laravel's Eloquent ORM. This is a pretty precise and effective approach. It makes sure that all the necessary data is fetched in a minimal number of queries, which consistently boosts performance across your application.We will also check the performance improvement using Laravel Debugbar, a tool we installed earlier in our environment via Composer, and confirmed through PhpStorm.We will be using our standard Linux development setup in "myapp" to execute all the steps.

Identify N+1 Query Problem

Here's some code you might want to consider. It'll get all the posts and then loop over each to access the comments: